Implementation of a hierarchical deep learning neural network for calculating finite element interpolations

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Objectives

In this research, neural networks informed by physics of the multi-layer perceptron type are used to improve the finite element method applied in an illustrative context of solid mechanics and to analyze, to a certain extent, the accuracy of the solutions found and the influence of parameters relevant to the configuration of the method.

Inspired by the articles Saha et al. (2021) and Zhang et al. (2021), the project aims to formulate and test a method that uses a hierarchical neural network to feedback the location of the nodes of a finite element mesh.

This feedback would come from the training of the network and, from there, it is expected that the finite elements will become more refined in the regions of greatest sensitivity of the solution, that is, the nodes would concentrate in the regions where the solution fluctuates the most, in order to reduce its error, while more linear regions of the solution would not concentrate so many elements.

Materials and Methods

The main neural network models and machine learning methods were introduced to the student through the courses SCC0230 - Artificial Intelligence and SCC0270 - Neural Networks and Deep Learning, taught, respectively, by Prof. Solange Oliveira Rezende and Prof. Moacir Antonelli Ponti.

Subsequently, PINNs (Physics-Informed Neural Network) were studied in greater depth based on articles such as Sharma et al. (2023) and video lessons on the internet. The finite element method was studied mainly based on Becker et al. (1981).

A first version of the algorithm was written in Python, using the PyTorch library, focused on machine learning, as well as other auxiliary libraries. However, after some difficulties in using the library, especially in its autograd package, the implementation of the algorithm was migrated to Matlab, where the network training was done manually, without the use of other libraries.

To formulate the method, the hat shape function and neural networks in parallel were considered for each of the nodes whose outputs are the analogue of this shape function. Thus:

$$u_h(x) = \sum_{i=1}^{N} u_I \phi_i(x) = \sum_{I=1}^{n} \mathcal{N}(\mathbf{W}, \mathbf{b}, \mathcal{A}, x) u_I$$

Results

The problem for applying the model was one of the examples in the article Zhang et al. (2021). In this example, there is a bar in a horizontal position, clamped at both ends and subjected to a vertical force of b(x):

$$b(x) = -\frac{4\pi^2(x-2.5)^2 - 2\pi}{e^{\pi(x-2.5)^2}} - \frac{8\pi^2(x-7.5)^2 - 4\pi}{e^{\pi(x-7.5)^2}}$$

After training the network, the position of the mesh nodes compared to the exact solution became as follows:

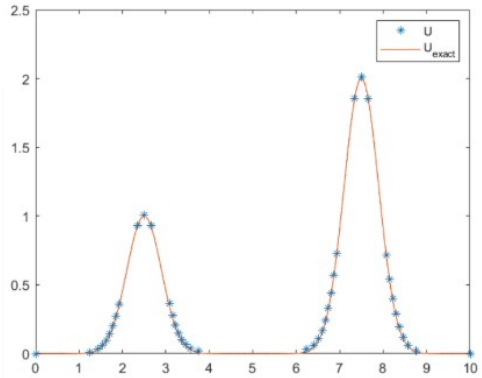


Figure 1: Positions of the nodes at the end of the training

Furthermore, the impact of changes in the parameters of the optimizer used to train the network on the model's learning was analyzed.

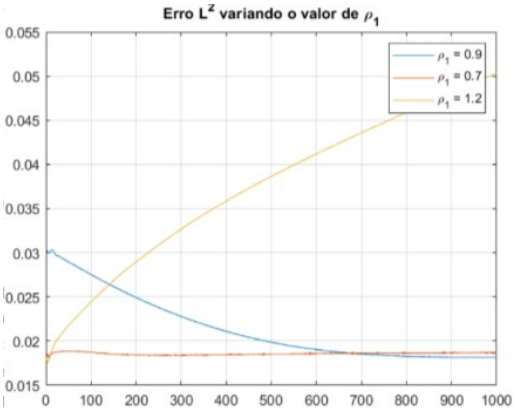


Figure 2: L² error as a function of one of the optimizer parameters

Conclusions

The model studied can in fact find a solution close to the exact solution and change the positions of the nodes as desired.

The study shows that the model analyzed reflects the great versatility of the neural network methodology, which can work well even in a context not related to collecting large databases, but only using mathematical equations governed by physics.

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